

Lab - V
Section: F
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Extra Questions

1. SR Flip-Flop:

Code:

```
module srff (input wire s, input wire r, input wire clk, input wire rst, output reg q );  
    always @(negedge clk)  
    begin  
        begin case ({s, r})  
            2'b00: q <= q;           // No change  
            2'b01: q <= 1'b0;       // Reset  
            2'b10: q <= 1'b1;       // Set  
            2'b11: q <= 1'bx;       // Don't Care  
        endcase  
    end  
end  
endmodule
```

```

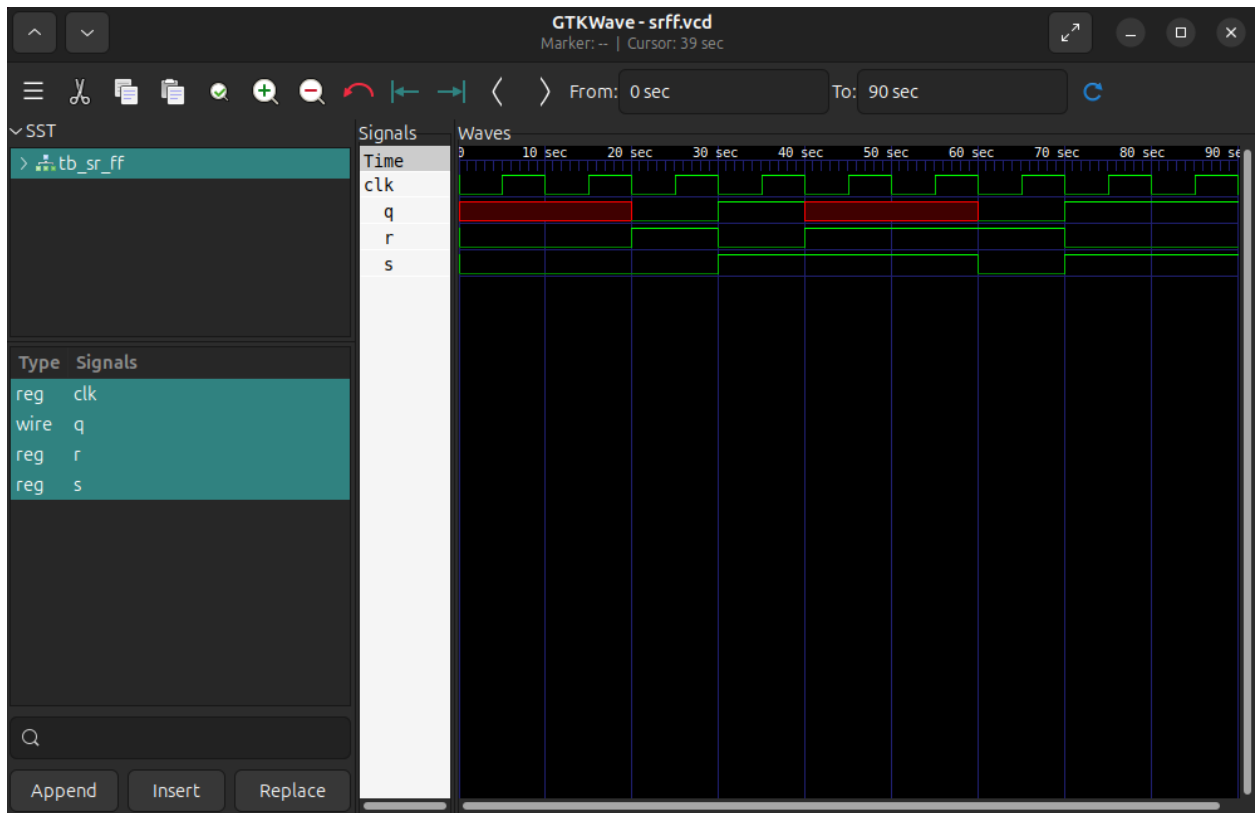
module tb_sr_ff;
reg s, r, clk;
wire q;
    // Instantiate JK flip-flop
    srff sr1 (.s(s),.r(r),.clk(clk),.q(q));
    // Generate clock
initial begin
    clk = 0;
forever #5 clk = ~clk;
end
initial
begin
    // waveform dump
    $dumpfile("srff.vcd");
    $dumpvars(0, tb_sr_ff);
end
    // Test sequence
initial begin           // Monitor outputs
    $monitor("Time=%0t | S=%b R=%b | Q=%b", $time, s, r, q);
    // Apply inputs on different clock edges
    #0 s = 0; r = 0;
    #10 s = 0; r = 0;    // Hold
    #10 s = 0; r = 1;    // Reset
    #10 s = 1; r = 0;    // Set
    #10 s = 1; r = 1;    // Don't Care
    #10 s = 1; r = 1;    // Don't Care
    #10 s = 0; r = 1;    // Reset
    #10 s = 1; r = 0;    // Set
    #20 $finish;
end
endmodule

```

VVP:

```
C0/LAB/LAB-V/SR_Flop_Top$ vvp srff
VCD info: dumpfile srff.vcd opened for output.
Time=0 | S=0 R=0 | Q=x
Time=20 | S=0 R=1 | Q=0
Time=30 | S=1 R=0 | Q=1
Time=40 | S=1 R=1 | Q=x
Time=60 | S=0 R=1 | Q=0
Time=70 | S=1 R=0 | Q=1
srff_tb.v:29: $finish called at 90 (1s)
```

GTKWave:



2. D Flip-Flop Using T Flip-Flop:

Code:

```
module tff (input wire T, input wire clk, output reg Q);
initial Q = 0;
always @(posedge clk )
    begin
        if (T)    Q <= ~Q;    // toggle
        else     Q <= Q;     // hold
    end
endmodule

module xorgate (y,a,b);
    input a,b;
    output y;
    assign y = a ^ b;
endmodule

module tfftodff (input wire D, input wire clk, output wire Q);
    wire T;
    xorgate x(T, D, Q);
    tff t1(T,clk, Q);
endmodule
```

```

module TB_D;
reg D;
reg CLK;
wire Q;
tfftodff newD (.D(D), .clk(CLK), .Q(Q));
    // clock
initial begin
    CLK = 1'b0;
    forever #5 CLK = ~CLK;
end
initial
begin
    // waveform dump
    $dumpfile("tfftodff.vcd");
    $dumpvars(0, TB_D);
end
initial
begin
    $monitor($time, " CLK=%b, D=%b, Q=%b", CLK, D, Q);
    D = 0;
    @(negedge CLK); D = 1;
    @(negedge CLK); D = 0;
    @(negedge CLK); D = 1;
    @(negedge CLK); D = 0;
    #20 $finish;
    $finish;
end
initial
begin
    #200 $finish;
end

```

VVP:

```
VCD info: dumpfile tfftodff.vcd opened for output.  
0 CLK=0, D=0, Q=0  
5 CLK=1, D=0, Q=0  
10 CLK=0, D=1, Q=0  
15 CLK=1, D=1, Q=1  
20 CLK=0, D=0, Q=1  
25 CLK=1, D=0, Q=0  
30 CLK=0, D=1, Q=0  
35 CLK=1, D=1, Q=1  
40 CLK=0, D=0, Q=1  
45 CLK=1, D=0, Q=0  
50 CLK=0, D=0, Q=0  
55 CLK=1, D=0, Q=0  
t_to_d_tb.v:25: $finish called at 60 (1s)  
60 CLK=0, D=0, Q=0
```

GTKWave:

